Università della Svizzera italiana | Faculty of Informatics

Bridging the Gap: Designing Decision-Support Tools for Small Hotel Management

Requirements Engineering, Design, and Evaluation of a Constraint-Solver Based Prototype

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Motivation

Every week, a hotel manager builds a staff schedule from scratch in Excel — checking availability, rest-day requests, occupancy, and coverage requirements entirely from memory. When asked whether dedicated software would help, the answer is almost always: *yes, but nothing suitable is available to me.*

Large chains use enterprise platforms costing thousands of euros per month. Independent hotels have no equivalent. Available tools either do too much (forced HR, payroll, POS bundles) or too little (a blank digital calendar). The algorithms exist; the end-user product does not.

This project builds that tool, starting from the requirements.

Requirements Elicitation

Analytical framework

6 expert interviews
hotel directors & general managers
independent properties
Ligurian Riviera

Tagging guide
10-category RE coding scheme
Beyer & Holtzblatt
Nuseibeh & Easterbrook

Transcription & tagging
line-by-line · 10 tag categories
AI-assisted cross-referencing

`CONSTRAINT-HARD`
`CONSTRAINT-SOFT`
`FEATURE-NEEDED` · `PAIN-POINT`
`DECISION-RULE` · `WORKFLOW`

~40 candidate features
clustered · ranked by importance
vs. implementation feasibility

20 features in scope
remaining deferred to future work

Design principles

Standalone first
no forced integrations

80/20 automation
system generates
manager approves

Hospitality-native shifts
split shifts · seasonal
non-standard weeks

Key findings

- **Excel universal** — rebuilt every week from scratch
- **Occupancy formulas** — sometimes already done manually in spreadsheets
- **Seniority pairing** — universal rule, never auto-enforced
- **Seasonal swings** — 10× staff variation is the norm
- **Payroll reconciliation** — hours by hand, every month
- **Full decision autonomy** — improvise or change prior decisions

Tool Survey

Category	Why SME hotels reject it
Integrated platforms	Prohibitive cost; all-or-nothing module adoption; opaque automation
Standalone SaaS	Generic shift model; no occupancy logic; setup overhead > time saved
Open-source solvers	Developer frameworks; no UI; no hospitality configuration

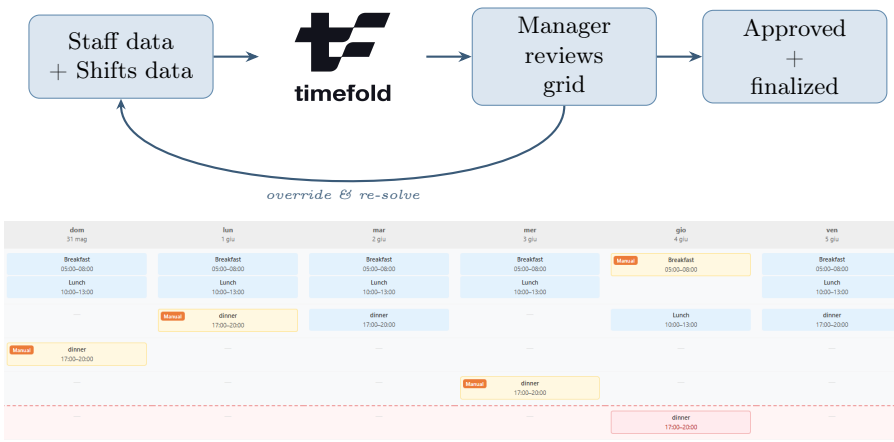
The gap is not a lack of technology — it is a mismatch.

System Architecture

Stack: React SPA → Spring Boot Back End → MongoDB Atlas;
Deployed on Render + Netlify.
Constraint solving via **Timefold Solver**;
AI extraction via **Spring AI + Anthropic Claude API**.

Key Features

Mixed-initiative scheduling loop



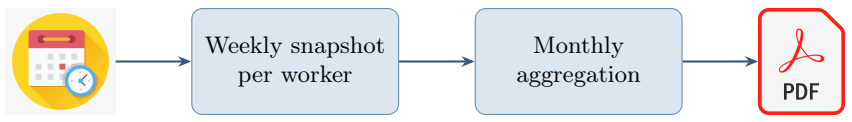
Constraint model

Hard constraints		Soft constraints	
×	No double assignment	~	Shift-type preferences respected
×	Weekly hours cap	~	Designated rest day honoured
×	Daily hours cap	~	Seniority balance per shift
×	Required headcount per shift slot	~	Manual overrides pinned

Occupancy engine



Monthly payroll report pipeline



Evaluation

5 hotel managers and directors completed a self-administered evaluation using the deployed prototype, following a structured nine-step walkthrough without researcher involvement or guidance.

System Usability Scale (SUS)

Mean SUS: 78.5 / 100

Good industry average: 68 excellent: >80.3

Feature relevance ratings

Feature	Mean
Occupancy-based headcount	**4.80**
Payroll report & PDF export	**4.80**
Manual override	**4.60**
Schedule auto-generation	**4.20**
AI-powered PMS import	**4.20**

Future Work

- Larger usability evaluation study
- Multi-tenant auth & roles
- Extended constraint model
- Direct PMS API integration